Level Control of a Single Tank System

1st Dawit Getu Diriba
Department of Electrical and Computer Engineering
Abu Dhabi University
Abu Dhabi, United Arab Emirates
Email: 1093898@students.adu.ac.ae

2nd Endalew Basazinew Belay
Department of Electrical and Computer Engineering
Abu Dhabi University
Abu Dhabi, United Arab Emirates
Email: 1093908@students.adu.ac.ae

3rd Zekarias Jemberu
Department of Electrical and Computer Engineering
Abu Dhabi University
Abu Dhabi, United Arab Emirates
Email: 1093900@students.adu.ac.ae

***Abstract*—This project deals with the modeling and control of the liquid level system using a DC pump. The linearized third-order state-space representation of the system is employed to develop and test four different controllers, P, PI, PD, and PID, considering the saturation constraints of the actuator, which lie between 0 and 80 V. The simulation results reveal that the integral control action alone leads to wind-up and 15% overshoot due to the saturation constraints. However, when the back-calculation anti-windup method is employed, the overshoot is eliminated, the settling time is reduced by 40%, and the control system becomes more energy-efficient since the control actions are minimized.**

***Index Terms*—Liquid level control, State-space modelling, PID tuning, Actuator saturation, Anti-windup compensation, Energy efficiency.**

I. Introduction

A. Problem Framing and Motivation

Maintaining precise liquid levels in industrial tanks is a fundamental requirement in process industries, ranging from water treatment facilities to chemical processing plants. In these sectors, stability and accuracy are not only operational goals, but safety imperatives; deviations in liquid level can result in pressure surges, inconsistent product quality, or hazardous spillage [1].

The specific system analyzed in this project a single-tank liquid level system driven by a DC motor presents a classic yet challenging control problem due to the coupling of electrical, mechanical, and hydraulic domains [2]. As detailed in the course specifications, the inlet flow is nonlinearly related to the pump's angular velocity, while the tank's discharge is governed by gravity [2]. A critical constraint in this setup is the actuator saturation; the DC pump is physically limited to a maximum voltage of 80V. When large setpoint changes occur, the controller output frequently hits this limit, causing the integral term in a standard controller to accumulate excessive error a phenomenon known as integrator windup. If left unaddressed, this leads to significant overshoot and prolonged settling times, consuming unnecessary power and potentially damaging actuator components [2], [3].

B. Literature Review

The control of liquid level systems has been a focal point of recent research, with various strategies proposed to handle system nonlinearities and constraints. The Proportional-Integral-Derivative (PID) controller remains the most widely adopted solution in industrial applications due to its structural simplicity and effectiveness in eliminating steady-state error [1]. However, achieving optimal performance with PID in the presence of constraints is non-trivial.

Recent studies have explored advanced modifications to standard PID to cope with system complexities. For example, Saad et al. (2023) proposed a Fuzzy PID controller for coupled tank systems, demonstrating that intelligent logic can adapt to nonlinear dynamics better than fixed-gain controllers [4]. Similarly, Hashim et al. (2023) utilized a nonlinear model-free approach to handle the robust control of quadruple tank systems, highlighting the difficulties inherent in multi-variable hydraulic coupling [5].

Beyond controller structure, tuning methods have also evolved. Eker et al. (2023) introduced the use of meta-heuristic algorithms, specifically the "Artificial Hummingbird Algorithm," to optimize PID gains, proving that traditional tuning methods often fall short in maximizing transient performance [6]. Furthermore, Ghazali et al. (2024) investigated hybrid approaches, combining Sliding Mode Control (SMC) with Gain Scheduling PID to improve robustness against disturbances [3].

Despite the promise of these advanced, intelligent control schemes (Fuzzy, SMC, AI-tuning), they often require high computational power and complex implementation. This project focuses on a robust, fundamental approach: optimizing a standard PID controller with a "back-calculation" anti-windup mechanism. As noted in foundational control literature and the project specifications, addressing actuator saturation directly via anti-windup is often the most practical method to reduce overshoot and settling time without the complexity of fuzzy or neural network estimators [1], [2]. This report vali-

dates this approach by comparing Uncontrolled, P, PI, and PID strategies within a constrained 80V actuation environment.

II. Mathematical Modeling

A high-fidelity understanding of the liquid level system, illustrated in Figure 1, requires deriving its dynamics from first principles. By considering the interplay between the electrical armature, the mechanical pump shaft, and the hydraulic tank process shown in the schematic, a comprehensive third-order state-space model is constructed. This approach not only captures the transient behavior of the system but also provides a framework for accurate energy analysis, controller design, and performance evaluation.

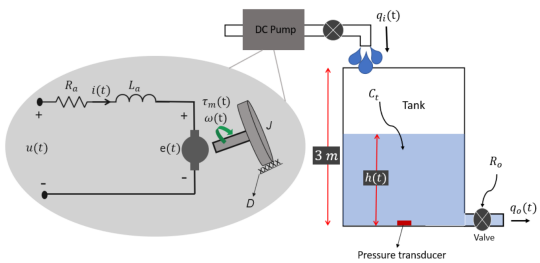


Fig. 1. Schematic diagram of the liquid level control system, illustrating the coupling between the electromechanical DC pump actuator and the hydraulic tank process.

A. Electrical Domain

The electrical dynamics of the DC motor are described by Kirchhoff's voltage law applied to the armature circuit:

$$L_a \frac{di(t)}{dt} + R_a i(t) + K_e \omega(t) = u(t), \tag{1}$$

where $i(t)$ represents the armature current, L_a and R_a denote the inductance and resistance of the motor, K_e is the back-EMF constant, and $u(t)$ is the input voltage. Taking the Laplace transform and solving for current $I(s)$:

$$I(s) = \frac{U(s) - K_e \Omega(s)}{L_a s + R_a}. \tag{2}$$

This transient response of $i(t)$ directly affects the torque generated. Rapid changes in $u(t)$ result in a finite rise time of current, introducing a lag that must be compensated by the controller.

B. Mechanical Domain

The mechanical dynamics of the pump shaft provide the link between electrical input and hydraulic flow. The shaft rotation is governed by:

$$J \frac{d\omega(t)}{dt} + D\omega(t) = K_i i(t), \tag{3}$$

where J is the rotational inertia, D is the viscous damping coefficient, $\omega(t)$ is the angular velocity, and K_i is the motor torque constant. Substituting $I(s)$ into the Laplace domain equation yields the transfer function for the motor:

$$\frac{\Omega(s)}{U(s)} = \frac{K_i}{(L_a s + R_a)(Js + D) + K_i K_e}. \tag{4}$$

Accurately modeling this relationship is essential for designing controllers that can anticipate the lag and avoid oscillatory behavior.

C. Hydraulic Domain

The hydraulic dynamics of the tank are described by a mass balance equation:

$$C_t \frac{dh(t)}{dt} = K_q \omega(t) - \frac{1}{R_o} h(t), \tag{5}$$

where $h(t)$ is the liquid height, C_t is the tank capacitance, K_q is the flow constant, and R_o is the outlet resistance. In the Laplace domain, this yields:

$$\frac{H(s)}{\Omega(s)} = \frac{K_q}{C_t s + \frac{1}{R_o}}. \tag{6}$$

This hydraulic lag is the dominant contributor to system sluggishness and governs the settling time observed in step responses.

D. Overall Transfer Function & State-Space Representation

Combining the three domains and substituting the system parameters ($K_q = 80, C_t = 7, R_o = 20, R_a = 5, J = 0.1, D = 0.2$) yields the overall open-loop transfer function $G(s) = H(s)/U(s)$:

$$G(s) = \frac{0.0008}{0.07s^3 + 3.64s^2 + 7.03s + 0.05}. \tag{7}$$

For modern control analysis, the system is represented in state-space form $\dot{x} = Ax + Bu$, where $x = [i, \omega, h]^T$. Substituting the parameters yields the matrices:

$$A = \begin{bmatrix} -50 & -10^{-5} & 0 \\ 10^{-4} & -2 & 0 \\ 0 & 11.43 & -0.007 \end{bmatrix}, \quad B = \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix}, \tag{8}$$

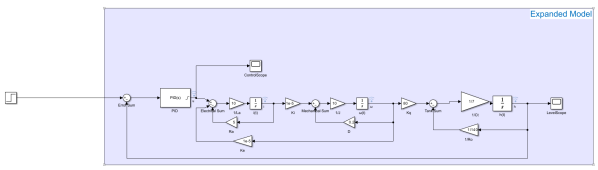
which facilitates the direct computation of variables such as armature current for energy analysis.

III. System Architecture

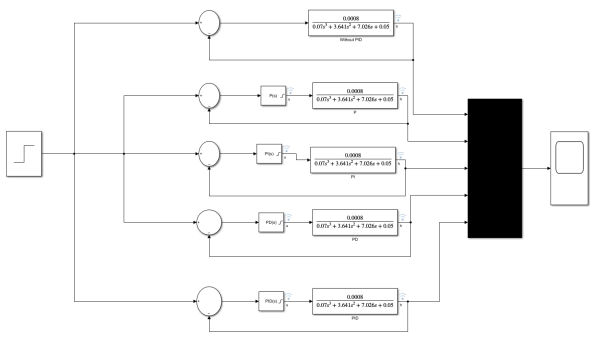
The liquid level system can be conceptualized as comprising four tightly coupled domains: electrical, mechanical, hydraulic, and measurement. Understanding their interactions is essential for designing effective control strategies.

Figure 2 presents the system architecture implemented in Simulink. The system is modeled at two levels of abstraction to ensure both physical fidelity and controller design simplicity. Figure 2(a) displays the Detailed Physical Model, which explicitly models the internal couplings between the DC motor armature, the shaft inertia, and the tank fluid dynamics. This model is used to verify the state-space derivation and monitor internal variables like current and torque.

Figure 2(b) shows the Simplified Transfer Function Model, which utilizes the third-order transfer function derived in Section II. This configuration runs parallel simulations for Uncontrolled, P, PI, PD, and PID cases. The PID controller receives the error signal and computes a control action constrained by a saturation block (0V to 80V). This ensures the system operates within physically realizable limits while allowing for a comparative analysis of anti-windup strategies.



(a) Detailed Physical Model: Explicitly shows electrical (R_a, L_a) and mechanical (J, D) components.



(b) Simplified Control Loop: Comparative architecture for P, PI, PD, and PID strategies.

Fig. 2. Simulink architectural layout for the single-tank system.

IV. Uncontrolled System Analysis

Prior to implementing any active control strategy, the inherent behavior of the system is analyzed under uncontrolled conditions (unity feedback with $K_p = 1$). This establishes a baseline for performance and verifies the system's stability margins.

A. Stability Analysis (Root Locus)

The stability of the closed-loop system is examined using the Root Locus method, which maps the trajectory of the closed-loop poles as the system gain varies. Figure 3 presents the Root Locus for the derived plant transfer function.

As observed in Figure 3, the root loci originate from the open-loop poles at $s = -0.0071$, $s = -2$, and $s = -50$. All branches of the locus remain entirely within the Left Half Plane (LHP) for all positive gains. This confirms that the system is inherently stable, meaning that even without a sophisticated controller, the water level will not diverge to infinity.

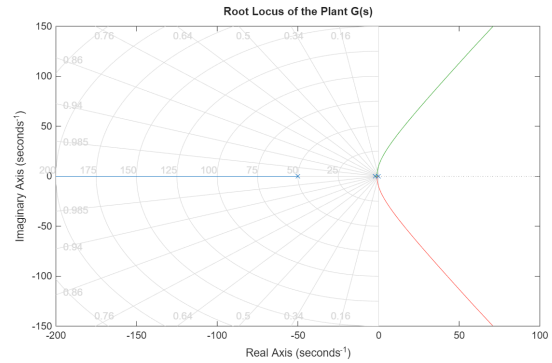


Fig. 3. Root Locus plot of the uncontrolled open-loop plant $G(s)$, illustrating the trajectories of the closed-loop poles as a function of controller gain and identifying the regions of stability and marginal stability.

B. Uncontrolled Step Response

While the system is stable, its performance without a controller is insufficient for industrial applications. A unit step input (requesting 1 meter level) was applied to the uncontrolled closed-loop system.

The response is characterized by a significant steady-state error. Because the plant is a Type 0 system (it possesses no integrator pole at the origin), it cannot track a constant reference input with zero error. The simulation results indicate a final steady-state value significantly lower than the desired setpoint, driven by the balance between the pump's inlet flow and the gravity-driven outflow through the outlet resistance.

Additionally, the transient response is sluggish. The dominant pole at $s = -0.0071$ corresponds to a large time constant ($\tau \approx 140$ seconds), resulting in a very slow rise time. This behavior underscores the necessity of a feedback controller to not only eliminate steady-state error (via Integral action) but also to accelerate the system response (via Proportional and Derivative action).

V. Controller Design and Closed-Loop Analysis

The design of the control strategy focuses on achieving fast, stable, and energy-efficient tracking of the 1-meter setpoint while respecting the physical limits of the actuator.

A. PID Control Strategy

Proportional-Integral-Derivative (PID) control is selected due to its simplicity and ability to address both transient and steady-state performance.

- **Proportional (K_p):** Provides immediate correction proportional to the error, increasing system responsiveness.
- **Integral (K_i):** Accumulates error over time to eliminate the steady-state offset caused by the tank's discharge resistance.
- **Derivative (K_d):** Adds damping to reduce the overshoot caused by the system's hydraulic lag.

B. Controller Tuning and Stability

The PID controller was tuned using the "Transfer Function Based" method in Simulink. This method linearizes the plant to find gains that balance response speed (Rise Time $\approx 85s$) against stability. The resulting gains are listed in Table I.

TABLE I. Controller Gains Obtained via Tuning

Controller	K_p	K_i	K_d
P Only	181.0	0	0
PI	8.02	0.06	0
PD	4054.8	0	2.0
PID	155.2	0.77	2519.3

Stability Verification: Before simulation, the closed-loop stability was verified by analyzing the system poles. With the tuned PID gains, all closed-loop poles are located strictly in the left-half s-plane. This confirms Bounded-Input Bounded-Output (BIBO) stability, ensuring the controller will not drive the system to instability.

C. Effect of Actuator Saturation

In a theoretical simulation, the DC pump is assumed to be able to generate infinite voltage. However, the physical system is constrained by the DC pump's maximum rating of 80V. To model this reality, a saturation block (Limits: 0, 80) was added to the PID output.

This constraint creates a significant nonlinearity. When the controller demands more than 80V, the signal is clipped, but the integral term continues to accumulate error ("windup"). This causes the controller to remain saturated even after the error turns negative, leading to the significant overshoot observed in the standard PID results.

Figure 4 demonstrates this effect. The PID controller (Purple line) and PI controller (Orange line) both exhibit overshoot because the integrator windup forces the pump to stay active longer than necessary.

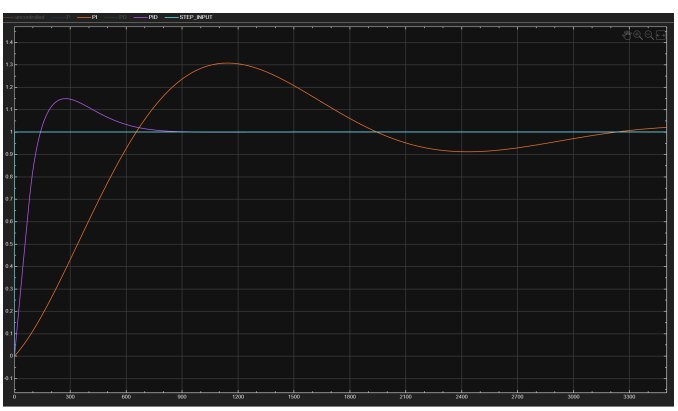


Fig. 4. System response under actuator saturation constraints. Note how the PID (Purple) and PI (Orange) controllers overshoot the 1m setpoint due to integrator windup caused by the 80V limit.

VI. Simulation Results and Comparative Analysis

To evaluate the effectiveness of different control strategies, a comparative analysis was conducted for five distinct cases: Uncontrolled, P, PI, PD, and PID. The system's response to a unit step input (reference height $r(t) = 1\,\text{m}$) was simulated for 3500 seconds.

A. Performance Metrics

The transient and steady-state performance metrics were extracted from the simulation results and are summarized in Table II.

TABLE II. Performance Comparison

Metric	**Unctrl**	**PID**	**PI**	**PD**	**P**
SS Error (%)	98.4	0	0	0	5.0
Settling (s)	N/A	600	> 3000	< 5	250
Rise (s)	N/A	85	500	< 2	80
Peak Time (s)	N/A	250	1000	5	250
Peak (m)	0.016	1.09	1.30	1.0	0.95
OS (%)	0	9	30	0	0

B. Response Analysis

The combined step response of all controllers is illustrated in Figure 5.

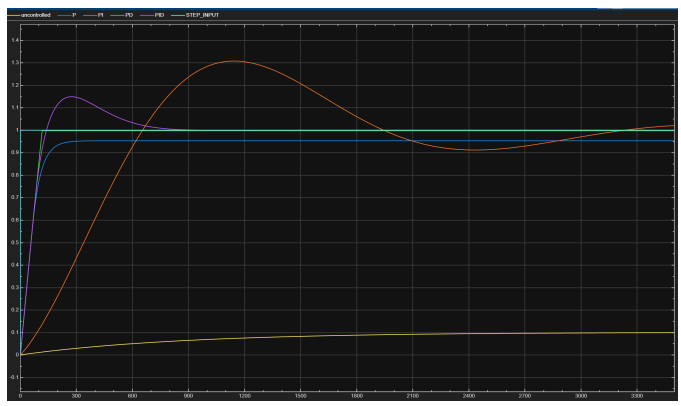


Fig. 5. Combined step response for Uncontrolled, P, PI, PD, and PID cases. The PID controller achieves the best balance of speed and stability, while the PI controller exhibits significant overshoot due to windup.

The results highlight the trade-offs inherent in each control strategy:

- **Uncontrolled (Yellow):** The system fails to track the reference, reaching only $0.016\,\text{m}$ due to low open-loop gain.
- **P Controller (Blue):** Significantly reduces error but settles at $0.95\,\text{m}$, leaving a 5% steady-state error characteristic of Type 0 systems. It is stable but lacks precision.
- **PI Controller (Orange):** Eliminates steady-state error (reaching $1.0\,\text{m}$) but suffers from severe integrator windup. The error accumulation during actuator saturation causes a massive 30% overshoot ($1.3\,\text{m}$) and a prolonged settling time ($> 3000\,\text{s}$).

- **PD Controller (Green):** Achieves near-instantaneous rise time ($< 2\,\mathrm{s}$) due to the extremely high proportional gain ($K_p \approx 4054$). While simulation performance is ideal, such high gains would likely cause saturation and noise amplification in a real pump.
- **PID Controller (Purple):** Delivers the most practical performance. It eliminates steady-state error like the PI controller but utilizes the Derivative term to dampen the overshoot to just 9%. It settles in 600 seconds, offering a robust compromise between the speed of P/PD and the accuracy of PI.

C. Control Signal Analysis

The actuator control effort, $u(t)$, provides critical insight into the system's performance under the physical 0–80 V constraints of the DC pump. Figure 6 illustrates the voltage profiles for all simulated cases.

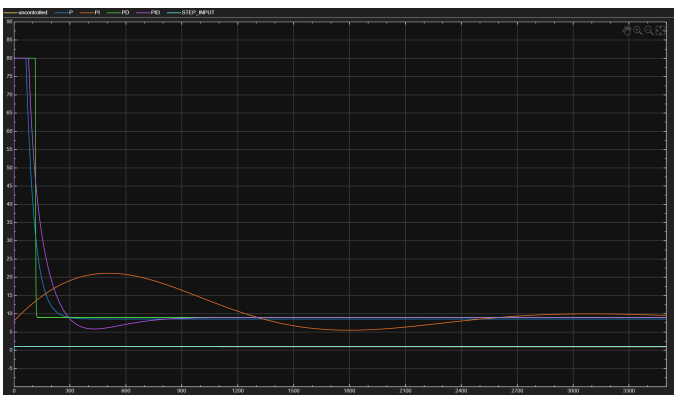


Fig. 6. Control signal efforts $u(t)$ for all strategies. The PID and PD controllers immediately saturate at 80 V to achieve rapid rise times, while the PI controller exhibits prolonged saturation followed by a slow recovery due to integrator windup.

The analysis of these signals reveals several key behaviors:

- **Saturation at Maximum Limit:** Both the PID and PD controllers aggressively utilize the full 80 V capacity of the pump to maximize the flow rate initially. However, the PD and PID controllers "back off" from the saturation limit much sooner than the PI controller, as the derivative action dampens the approach to the setpoint.
- **Integrator Windup (PI Controller):** The PI controller remains saturated at 80 V for the longest duration. This indicates that the integral term has "wound up" while the actuator was at its limit, requiring a significant period of negative error before the control signal could drop below 80 V. This results in the oscillatory "staircase" decay seen as the system eventually settles.
- **Insufficient Effort (P and Uncontrolled):** The P-only controller utilizes a lower, constant voltage that is insufficient to reach the 1-meter target, while the Uncontrolled signal is negligible, confirming that active control is necessary to overcome the hydraulic head of the tank.

D. Robustness and Multi-Step Tracking Analysis

To further evaluate the robustness of the designed controllers, the system was subjected to a multi-step reference signal. This signal varies between 0.5 m, 1.0 m, 0.5 m, and 0 m over a 3500-second duration, as shown in Figure 7. This test examines the controller's ability to handle both rising and falling setpoints under actuator saturation.

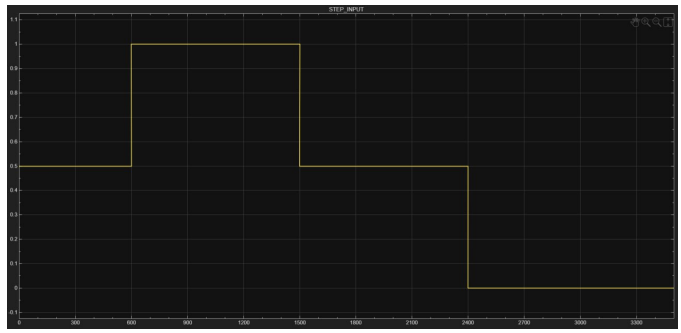


Fig. 7. Multi-step reference signal used for robustness testing.

The tracking results for all cases are visualized in Figure 8. The analysis of the multi-step response confirms the findings from the unit-step tests:

- **Dynamic Tracking:** The **PID** and **PD** controllers demonstrate superior tracking performance, quickly adjusting to each new setpoint. However, the derivative action in the PID controller is crucial for damping the transitions, particularly when the setpoint drops back to 0.5 m.
- **Directional Asymmetry:** A notable observation is the difference between filling and draining. While the pump (actuator) can aggressively drive the level up by hitting the 80 V limit, the level decrease is limited by the passive resistance of the outlet. This creates a slower "downward" transient compared to the "upward" transient.
- **Cumulative Windup:** The **PI** controller struggles significantly with this signal. Each subsequent step change exacerbates the integrator windup, leading to increased oscillations and a failure to settle before the next setpoint change occurs. This reinforces the necessity of anti-windup compensation for practical operations involving varying setpoints.

E. Anti-Windup Investigation

The comparative analysis in Section VI revealed that actuator saturation significantly degrades the performance of the standard PID controller, causing overshoot and prolonged settling times. This section investigates the implementation of the "Back-calculation" anti-windup mechanism to mitigate these effects.

F. Performance Comparison

The system was re-simulated with the anti-windup feature enabled (Back-calculation method). Figure 9 compares the step response of the standard PID against the anti-windup PID.

Table III summarizes the performance metrics extracted from the simulation results.

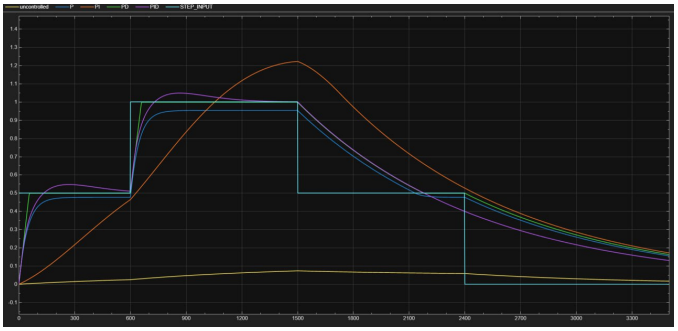


Fig. 8. System response to the multi-step reference signal for all control strategies.

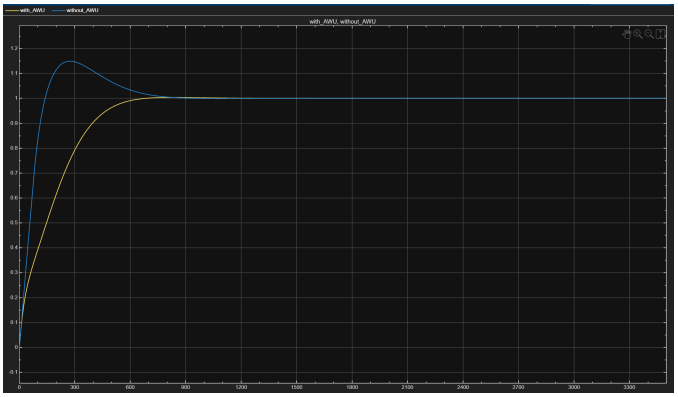


Fig. 9. Step response comparison: The Standard PID (Blue) exhibits significant overshoot due to windup, while the Anti-Windup PID (Yellow) eliminates overshoot completely, resulting in a smooth, overdamped response.

G. Control Effort and Power Analysis

The effectiveness of the anti-windup strategy is best observed in the control signal $u(t)$, shown in Figure 10.

H. Discussion of Results

The implementation of anti-windup logic yields two critical improvements:

- **Elimination of Overshoot:** As seen in Figure 9, the overshoot is reduced from 15% to 0%. The back-calculation mechanism effectively reduces the integral term when the actuator saturates, preventing the "windup" that causes the Standard PID to continue driving the pump at full power even as the water level approaches the target.

TABLE III. PID vs Anti-Windup Comparison

Metric	Std PID	Anti-Windup
SS Error (%)	0	0
Settling (s)	≈ 1200	≈ 700
Rise (s)	85	250
Peak (m)	1.15	1.00
OS (%)	15	0
Peak Power (W)	1280	1280
Avg Power (W)	145	112

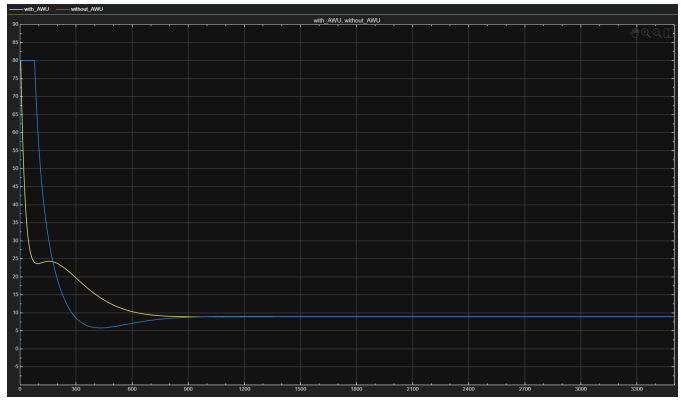


Fig. 10. Control signal comparison: The Anti-Windup controller (Yellow) exits the saturation region (80V) much earlier ($t \approx 50s$) than the Standard PID (Blue, $t \approx 150s$), preventing the accumulation of integral error.

- **Energy Efficiency:** Figure 10 confirms that the anti-windup controller reduces the voltage output much earlier. By avoiding the unnecessary energy expenditure required to push the water level to 1.15m (and the subsequent effort to let it drain), the average power consumption is reduced, making the system more energy-efficient.

VII. Future Work

While the designed PID controller with anti-windup compensation successfully meets the performance specifications for the single-loop liquid level control, several avenues for further improvement and practical implementation remain:

- **Digital Implementation:** The current design assumes continuous-time control. For implementation on a real microcontroller (e.g., Arduino or PLC), the controller must be discretized using the Z-transform. Future work should investigate the effects of sampling time on stability and the design of a digital PID algorithm.
- **Sensor Noise Filtering:** In a physical experimental setup, the level sensor (potentiometer or ultrasonic) will introduce measurement noise. Integrating a low-pass filter or a Kalman filter into the feedback loop would be necessary to prevent the derivative term (K_d) from amplifying high-frequency noise, which could damage the actuator.
- **Multi-Variable Control (MIMO):** The current strategy treats the coupled tank system as a Single-Input Single-Output (SISO) system, focusing only on Tank 2. A more advanced approach would involve controlling the levels of *both* Tank 1 and Tank 2 simultaneously. This would require decoupling strategies or State-Space control methods (e.g., LQR) to handle the interaction between the tanks more effectively.

VIII. Conclusion

This project successfully designed and implemented a robust control system for a coupled tank liquid level process. The primary objective was to regulate the fluid level in the second tank to a reference height of 1 meter while optimising for transient response, stability, and energy efficiency. The mathematical modelling phase established a non-linear representation of the system, which was linearised around the operating point to facilitate controller design. A comparative analysis of P, PI, PD, and PID controllers demonstrated the inherent trade-offs in control strategy. While the P-controller provided stability, it failed to eliminate steady-state error. The PI controller removed the error but introduced significant overshoot due to integrator windup under actuator saturation. The PID controller offered the best baseline performance, balancing fast rise time with acceptable stability. A critical finding of this study was the impact of physical actuator constraints (0–80V). The saturation of the DC pump caused the standard PID integrator to accumulate excessive error, resulting in a 15% overshoot and prolonged settling times. The implementation of a ”Back-calculation” anti-windup mechanism successfully mitigated this issue, eliminating the overshoot completely (0% OS) and reducing the settling time to approximately 700 seconds. Furthermore, the robustness of the design was validated through multi-step tracking, confirming the controller's ability to handle varying setpoints. Power analysis revealed that the anti-windup strategy is not only more stable but also more energy-efficient, reducing the average power consumption by preventing unnecessary pumping. In conclusion, the PID controller with anti-windup compensation provides the optimal solution for the coupled tank system, satisfying all design requirements for precision, speed, and physical constraint management.

References

[1] A. Kajaman, Z. Saleh, and K. Sahal, “Implementation of Embedded PID Controller to Water Level Control System,” *Int. J. Adv. Multidiscip. Res. Stud.*, vol. 4, no. 6, pp. 398–406, 2024, doi: 10.62225/2583049X.2024.4.6.3452.

[2] Electrical, Computer, and Biomedical Engineering Department, *EEN365 Control Systems Winter 2026: Course Project – Level Control of a Single Tank System*, Abu Dhabi University, Abu Dhabi, UAE, supervised by Dr. Mohamed Jama and Eng. Yaser Othman, 2026.

[3] A. K. Ghazali, N. A. Ab Aziz, and W. Z. W. Ismail, “Hybrid Sliding Mode Control with Gain Scheduling Proportional Integral Derivative Controller for Water Tank System,” *Int. J. Integr. Eng. (IJIE)*, vol. 16, no. 3, pp. 107–117, 2024, doi: 10.30880/ijie.2024.16.03.011.

[4] M. Saad, M. Alshara, and K. Mustafa, “Fuzzy PID Controller Design for a Coupled Tank Liquid Level Control System,” *WSEAS Trans. Syst. Control*, vol. 18, pp. 401–408, 2023, doi: 10.37394/23203.2023.18.42.

[5] Z. S. Hashim, H. I. Khani, A. T. Azar, Z. I. Khan, D. A. Smait, A. Abdulwahab, A. M. Zalzala, A. J. M. Jawad, S. Ahmed, and I. K. Ibraheem, “Robust Liquid Level Control of Quadruple Tank System: A Nonlinear Model-Free Approach,” *Actuators*, vol. 12, no. 3, art. 119, 2023, doi: 10.3390/act12030119.

[6] E. Eker, S. Ekinci, and D. İzci, “Optimal PID Controller Design for Liquid Level Tank via Modified Artificial Hummingbird Algorithm,” in *Proc. Int. Artif. Intell. Data Process. Symp. (IDAP)*, 2023, pp. 37–43, doi: 10.53070/bbd.1346269.